

Gastrointestinal tract (GIT)

Enzymes and hormones

Stage 3

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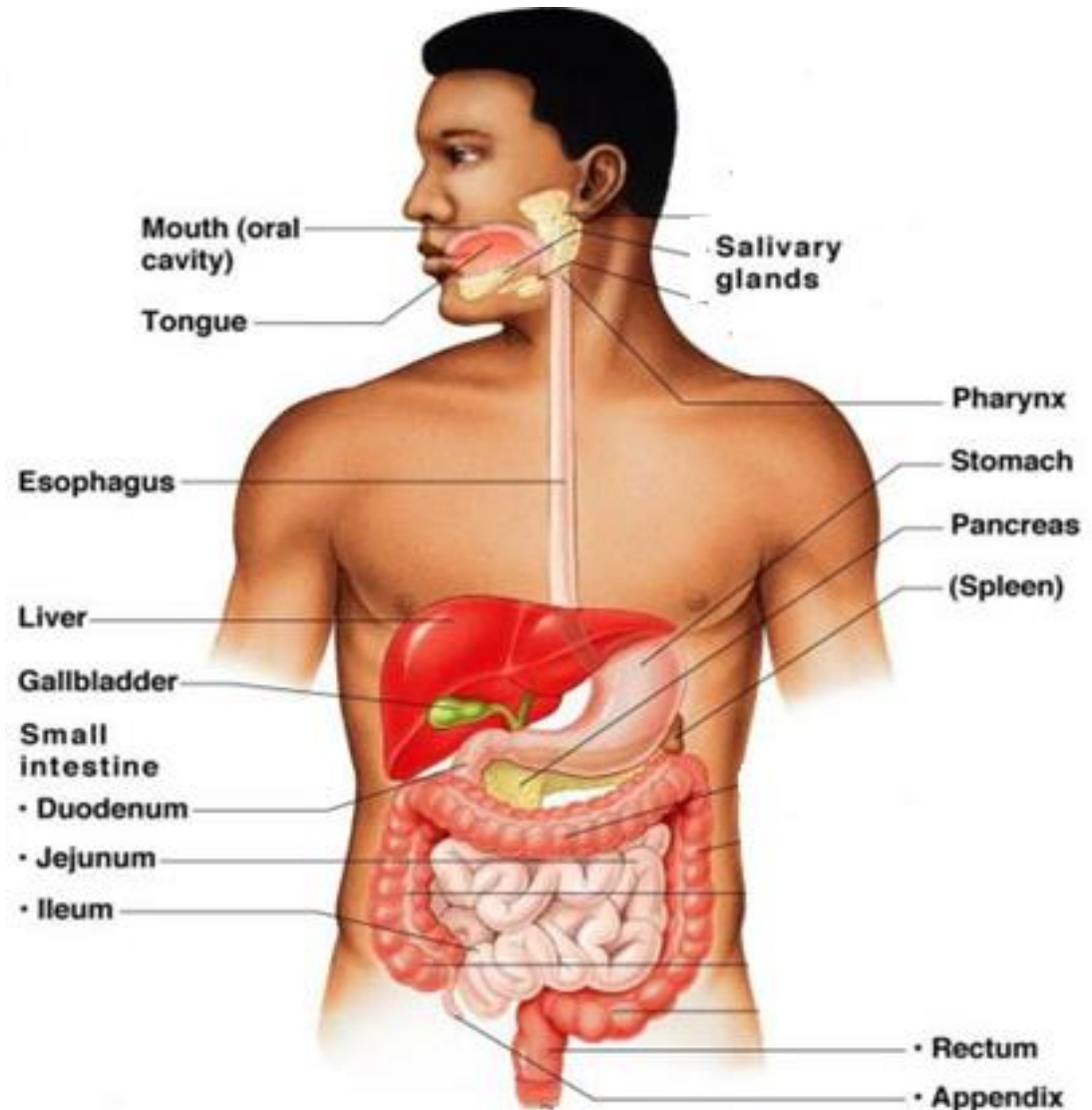
Digestive system

- Is a series of tube-like organs that convert foods into body energy
- The breakdown of **large insoluble** molecules into **smaller soluble** molecules which can pass through the wall of the gut into the blood.

The Human Digestive System

Main organs of the digestive system:

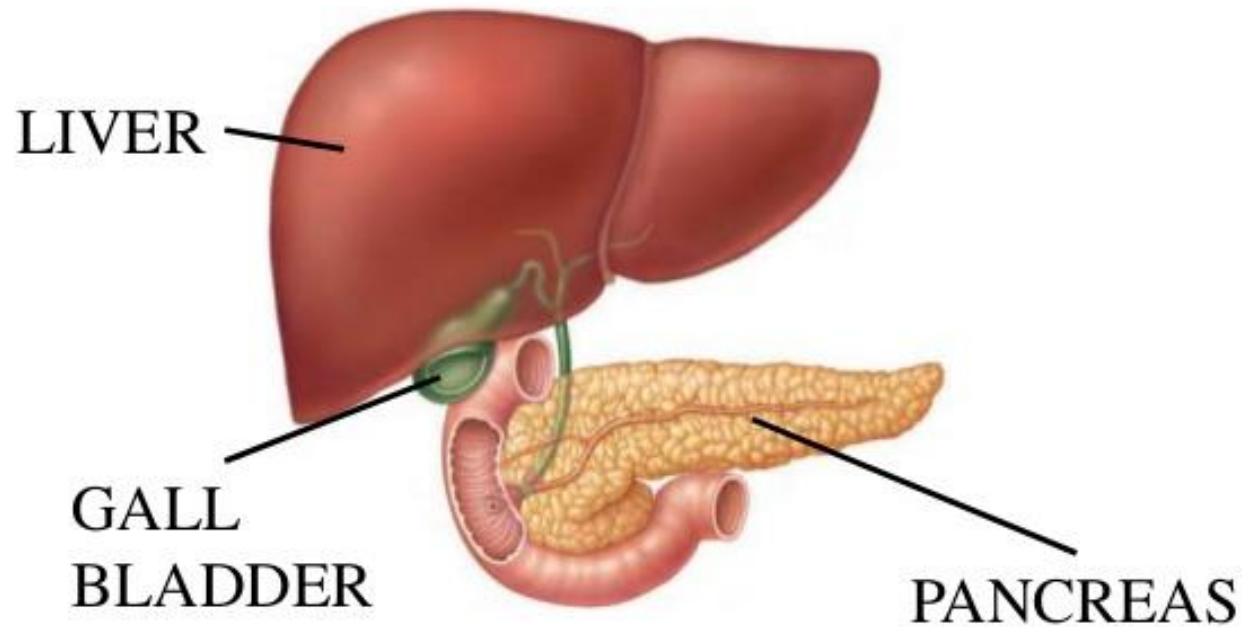
- Mouth
- Pharynx
- Esophagus
- Stomach
- Small intestine
- Large intestine.



Accessory organs of the Digestive System:

- **Teeth**
- **Tongue**
- **Liver**
- **Pancreas**
- **Gall Bladder**
- **Salivary Gland**
- **Note:** The accessory organs of the digestive system are not a part of the alimentary canal (food does not pass through them) but they assist the alimentary organs in the process of digestion

DIGESTIVE SYSTEM : ACCESSORY STRUCTURES



Functions of the digestive system:

- Ingestion.
- Secretion.
- Mixing & Propulsion.
- Digestion

a-Mechanical: Food is broken down mechanically, with movement and

b-Chemical: Food is broken down with a chemical (or enzyme)

- Absorption.
- Defecation

The GIT regulatory substances (Enzymes)

Enzymes:

- **Are biocatalysts in biological systems that speed up the rate of the biochemical reaction without being used up.**
- **Are essential to life because most chemical reactions in biological cells would occur too slowly or would lead to different products without enzymes.**

Functions & characteristics of enzymes

- Are water soluble highly specialized proteins
- Most enzymes are globular proteins
- Some special RNA species also act as enzymes & are called **Ribozymes***
- Enzymes speed up the reaction by lowering the activation energy of the reaction.
- Their presence does not affect the nature and properties of the end product.
- They are highly specific in their actions that are each enzyme can catalyze one kind of substrate.

*Ribozymes

- ❑ Ribozyme is derived from the word 'RNA' and 'enzyme'.
- ❑ Term ribozyme is used to describe an RNA with catalytic activity.
- ❑ Ribozyme activity is based on transesterification and phosphodiester bond hydrolysis.
- ❑ Ribozymes are inactivated by heating above their melting temperature and if essential nucleotides are changed.

Digestive Enzymes: Types and Function

- **Are substances that help you digest your food.**
- **Are secreted (released) by the salivary glands and cells lining the stomach, pancreas, & small intestine.**
- **Digestive enzymes do this by splitting the large, complex molecules that makeup proteins, carbs, and fats into smaller ones. This allows the nutrients from these foods to be easily absorbed into your blood and carried through your body.**

Types of Digestive Enzymes

The most important digestive enzymes are:

- **Amylase**
- **Maltase**
- **Lactase**
- **Sucrase**
- **Lipase**
- **Proteases**

Some foods require certain digestive enzymes to break down the specific nutrients they contain.

Amylases

Are a family of enzymes that catalyzes the hydrolysis of starch into dextran, maltose and glucose

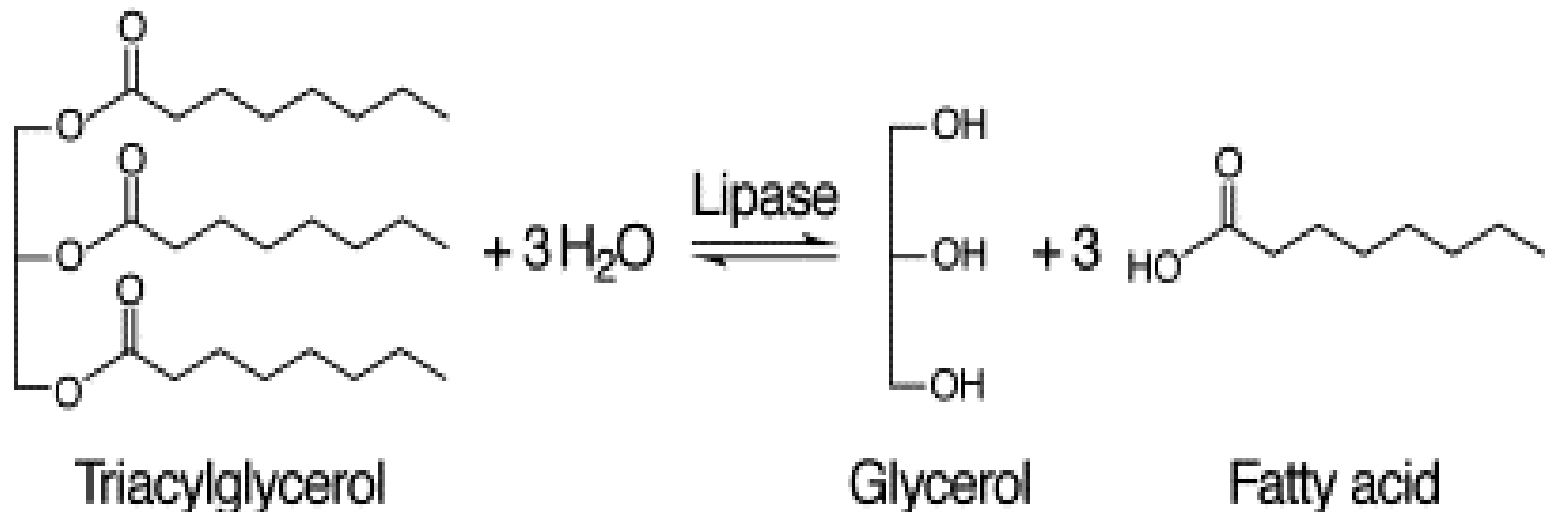
Source:

Present in a number of organs and tissues such as

- **Salivary gland,**
- **Pancreas,**
- **Weak amylase activity is found in**
- **Liver extracts,**
- **Lung,**
- **Fallopian tubes,**
- **Ovary & prostate**

Lipases

- Are a family of enzymes that catalyzes the hydrolysis of fats into fatty acids and glycerol
- It's produced in small amounts by your mouth and stomach, and in larger amounts by your pancreas.



Proteases

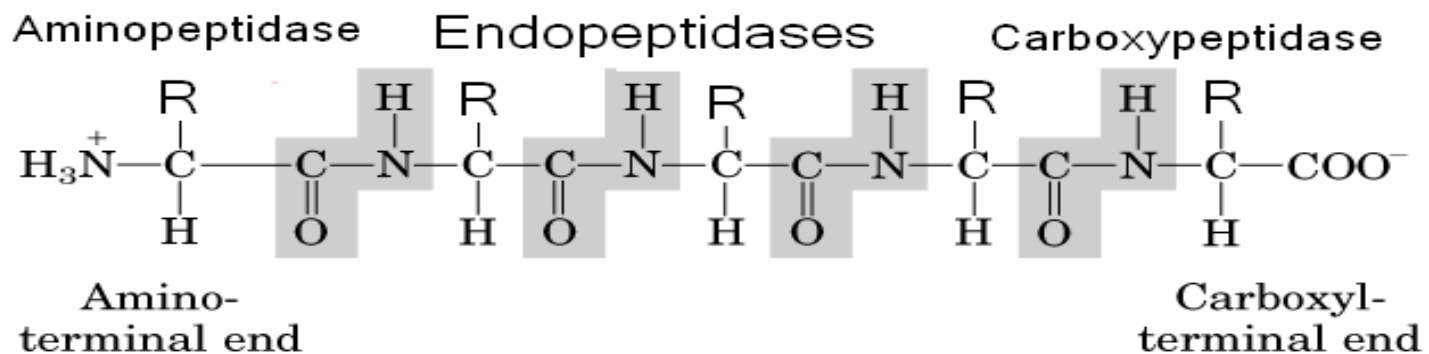
- **Also called peptidases, proteolytic enzymes, or proteinases,**
- **A protease is an enzyme that catalyzes proteolysis, breaking down proteins into smaller polypeptides or single AAs.**
- **Proteases are produced in the stomach and pancreas.**

The proteolytic enzymes of G.I.T. are of two types:

1-Endopeptidases: hydrolyze the inner peptide bonds of peptides or proteins.

2-Exopeptidases: subdivided into two subtypes:

- **Aminopeptidases:** hydrolyze the peptide bond of the N-terminal AA.
- **Carboxypeptidases:** hydrolyze the peptide bond of the C-terminal AA.



Pepsin:

- **Pepsin:** this acid stable endopeptidase is secreted as an inactive **zymogen** pepsinogen in the stomach.
- pepsinogen is activated to pepsin, either by HCl or auto-catalytically by other pepsin molecules that have already been activated,
- pepsin release peptides and a few free AAs from dietary protein
- It is specific for peptide bonds formed by aromatic or acidic AAs,

Rennin: in infants for digestion of milk protein (Casein).

Trypsin:

- Is an enzyme in the 1st section of the small intestine that starts the digestion of protein which is secreted by the pancreas,
- Trypsin then activates additional pancreatic enzymes, such as **carboxypeptidase** and **chymotrypsin**, to help break down peptides.
- Trypsin cleaves only when the carbonyl group of the peptide bond is contributed by arginine or lysine

- **Chymotrypsin**: This enzyme breaks down peptides into free amino acids that can be absorbed by the intestinal wall.

Chymotripsin ... Trp, Tyr, Phe, Met, leu

- **Carboxypeptidase A**: Secreted by the pancreas, it splits peptides into individual AAs.

Carboxypeptidase A, Ala, Ile, Leu, Val

- **Carboxypeptidase B**: Secreted by the pancreas, it breaks down basic amino acids

Carboxypeptidase B, Arg and Lys

- **Elastase** ...Ala, Gly, Ser

The diagram illustrates the activation of pancreatic enzymes and their substrate specificity. A polypeptide chain is shown with specific amino acids highlighted in parentheses: (Arg, Lys), (Trp, Tyr, Phe, Met, Leu), (Ala, Gly, Ser), and (Ala, Ile, Leu, Val) or (Arg, Lys). Below the chain, four enzymes are shown: Trypsin, Chymotrypsin, Elastase, and Carboxypeptidase A/B. Arrows indicate the activation of these enzymes from their precursors: Trypsinogen to Trypsin (via Enteropeptidase), Chymotrypsinogen to Chymotrypsin, Proelastase to Elastase, and Procarboxypeptidase A/B to Carboxypeptidase A/B. A dietary protein is shown entering the system.

Dietary protein

Trypsin

Chymotrypsin

Elastase

Carboxypeptidase A
Carboxypeptidase B

Enteropeptidase

Trypsinogen

Chymotrypsinogen

Proelastase

Procarboxypeptidase A
Procarboxypeptidase B

What causes enzyme deficiency?

Things that can cause an enzyme deficiency include:

- **Diet high in sugar and processed foods**
- **Repeated antibiotic use**
- **Smoking**
- **Aging**
- **Pregnancy**
- **Physical, psychological and emotional stress**

Eight signs you might have an enzyme deficiency:

- Food intolerances
- Abdominal pain or discomfort
- Diarrhea & Bloating
- Constipation & Undigested food in the stools
- Poor nutrient absorption
- Weaker immune system
- Fatigue, Anxiety & depression
- Skin problems

Metabolic disorders associated with Digestive Enzyme Deficiencies

There are a variety of health conditions that can interfere with the secretion of enough digestive enzymes to fully digest foods. Some are inherited genetic conditions while others develop over time.

Effect:

The presence of lactose in the intestine causes:

- **Increased osmotic pressure:** So water will be drawn from the tissue (causing dehydration) into the large intestine (causing diarrhea).
- **Increased fermentation of lactose by bacteria:** Intestinal bacteria ferment lactose with subsequent production of CO₂ gas. This causes abdominal cramps.
- **Treatment:** Treatment of this disorder is simply by removing lactose (milk) from the diet.

Exocrine Pancreatic Insufficiency

- **The pancreas produces the key digestive enzymes amylase, protease, and lipase.**
- **People with exocrine pancreatic insufficiency (EPI) have a deficiency of these enzymes. As a result, they are unable to digest food properly, especially fats.**

The health conditions that affect the pancreas and are associated with EPI are:

- **Chronic pancreatitis:** An inflammation of the pancreas that can permanently damage the organ over time
- **Cystic fibrosis:** An inherited genetic condition that causes severe damage to the lungs and digestive system, including the pancreas
- **Pancreatic cancer**

The GIT regulatory substances (Hormones)

Introduction

- **A hormone is a chemical substance released by a cell or a gland in one part of the body that sends out messages that affect cells in other parts of the organism.**
- **It is a chemical messenger that transports a signal from one cell to another.**

- Secreted in very small amounts
- Transported, usually, in the blood
- Target cells have specific receptors
- Regulates cell reactions by affecting gene expression (often gene transcription factors)
- Regulate the metabolic events

The GIT hormones:

- **Are secreted in GIT.**
- **Are belong to the family of local hormones**
- **It controls GI secretion and regulates the motility of the GIT.**
- **Most of them are peptides.**
- **Many of them are also found in other tissues, especially the brain.**

The GIT hormones:

1- Gastrin

2- Cholecystokinin

3- Secretin

4- Somatostatin

5- GIP

6- Ghrelin

1- Gastrin:

Is a peptide hormone released by the “ γ ” cells of the stomach, duodenum & pancreas in response to stimuli associated with the ingestion of a meal.

Stimuli for Release

Presence of peptides and AAs in the gastric lumen

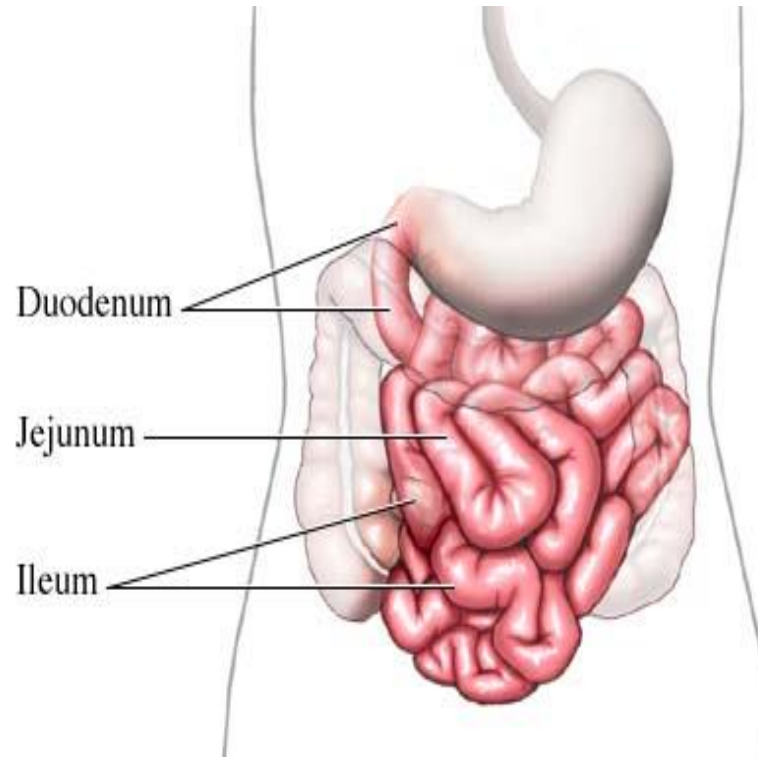
What are the main actions of gastrin?

The primary actions of gastrin are:

- It stimulates the exocrine cells of the stomach to secrete gastric juice (a mixture of HCl & the proteolytic pepsin enzyme)**
- Increased gastric motility**
- The proliferation of gastric epithelium**

2- Cholecystokinin (CCK):

Is a peptide hormone of the GIT system secreted by duodenum and jejunum mainly in response to digestive products of **fat, FAs, protein & monoglycerides** in the intestinal contents



What does Cholecystokinin do?

Action

- This hormone strongly contracts the gallbladder, expelling bile into the small intestine where the bile in turn plays important role in emulsifying fatty substances, allowing them to be digested and absorbed.**
- Also, it inhibits stomach contraction, & causes emptying of the gall bladder.**
- It also slows the emptying of food from the stomach.**
- Its effects on the pancreas stimulating the release of pancreatic digestive enzymes into the pancreatic fluid**

What stimulates the release of **Cholecystokinin?**

Stimuli for Release

Presence of fatty acids and amino acids in the small intestine

3- Secretin:

Was the first GI hormone discovered & is secreted by the duodenum & upper jejunum in response to acidic gastric juice emptying into the duodenum

Functions & Mode of secretin

- **Is a hormone that controls the secretions into the duodenum & also regulates water homeostasis throughout the body**
- **Stimulates pepsin secretion**
- **It reduces acid secretion from the stomach by inhibiting gastrin release from G cells**
- **Stimulates the secretion of bile.**
- **It regulates the pH of the duodenal contents via the control of gastric acid secretion and buffering with bicarbonate**

- **It also enhances the effect of cholecystokinin to induce the secretion of digestive enzymes and bile from pancreas and gallbladder respectively.**
- **It stimulates gallbladder contraction**
- **It stimulates the exocrine portion of the pancreas to secrete bicarbonate into the pancreatic fluid (thus neutralizing the acidity of the intestinal contents).**
- **Increased insulin release from the pancreas, following oral glucose intake.**

4-Somatostatin (GH release inhibitory factors)

- **Is a hormone that regulates a variety of bodily function**
- **Was first found in the hypothalamus.**
- **It is found in δ cells of the stomach and the upper part of the small intestine and δ cells of the pancreatic islet.**
- **Is secreted in two forms. One with 14 amino acids and the other one with 28 amino acids.**

Stimulant for secretion

- The presence of chyme with glucose and proteins in the stomach and small intestine causes the release of somatostatin.

What does Somatostatin do?

Action

- Inhibits the secretion of growth hormone and thyroid stimulating hormone (TSH) from the anterior pituitary.
- Inhibits gastric HCl secretion and motility
- Inhibits secretion of pancreatic juice
- Inhibits both insulin and glucagon secretion
- Inhibits secretion of GI hormones such as
 - (a) Gastrin
 - (b) CCK
 - (c) Motilin

5- GIP (gastric inhibitory peptide)

Is secreted by duodenum and jejunum mucosa

Action

- Inhibits gastric acid secretion & Gastric motility**

6- Ghrelin

- Is a lipopeptide consisting of 28 AAs with a covalently attached 8-carbon fatty acid.
- Is secreted by endocrine cells in the stomach, especially when one is hungry and
- Acts on the hypothalamus to stimulate feeding.
- This action counteracts the inhibition of feeding by leptin.